

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.10 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

- Q.1 The value of a for which all extrema of function $f(x) = x^3 - 3ax^2 + 3(a^2 - 1)x + 1$ lie in the interval $(-2, 4)$ is
(A) $(3, 4)$ (B) $(-1, 3)$ (C) $(-3, -1)$ (D) $(-2, -1)$
- Q.2 Let $f(x) = xe^x(1-x)$, $x \in \mathbb{R}$ then $f(x)$ is
(A) increasing on $\left[-\frac{1}{2}, \frac{1}{2}\right]$ (B) decreasing on $\mathbb{R}$
(C) increasing on $\mathbb{R}$ (D) decreasing on $\left[-\frac{1}{2}, 2\right]$
- Q.3 If $f(x) = x(x-2)(x-4)$, $1 \leq x \leq 4$, then a number satisfying the conditions of the lagrange's mean value theorem is
(A) 2 (B) 3 (C) $5/2$ (D) $7/2$
- Q.4 A function is continuous & differentiable on $\mathbb{R}_0$ and satisfies the condition $xf'(x) + f(x) = 1$ throughout its domain, with $f(1) = 2$. Then the range of the function is
(A) $(-\infty, \infty)$ (B) $(-\infty, 1) \cup (1, \infty)$ (C) $(0, \infty)$ (D) $(1, \infty)$
- Q.5 If $f(x)$ satisfies all condition of Rolle's theorem in $[2, 3]$, then $\int_2^3 f'(x) dx$ is equal to
(A) 0 (B) 1 (C) 2 (D) 3
- Q.6 If the tangent drawn to the curve $f(x) = -e^{-x}(x^2 + 2x + 2) + x$ at $(\alpha, f(\alpha))$ meets the curve again at two distinct points and the range of α is (a, b) then the value of $(a+b)$ is
(A) 1 (B) 2 (C) 4 (D) 5
- Q.7 Let $f(x) = \cot^{-1}(g(x))$, where $g(x)$ is monotonically decreasing for $0 < x < \frac{\pi}{4}$, then $f(x)$ is
(A) monotonically increasing in $\left(0, \frac{\pi}{4}\right)$ (B) monotonically decreasing in $\left(0, \frac{\pi}{2}\right)$
(C) non-monotonic in $\left(0, \frac{\pi}{4}\right)$ (D) monotonically increasing in $\left(0, \frac{\pi}{4}\right)$

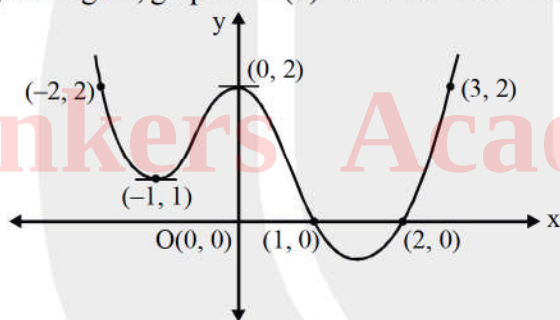
- Q.8 The function $f(x) = \int_0^{2x^2} (t-2)(t+1)^3(t-3)^2 dt$ have number of maxima = a and number of minima = b then (a + b) is
 (A) 2 (B) 3 (C) 4 (D) 5
- Q.9 If an equation of a tangent to the curve $y = \cos(x+y)$, $-1 \leq x \leq 1 + \pi$, is $x + 2y = k$, then $k =$
 (A) 1 (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) 2
- Q.10 If the maximum value of the function $f(x) = \sqrt{x^4 - 3x^2 - 6x + 13} - \sqrt{x^4 - x^2 + 1}$ is $\sqrt{M}$ for the permissible real x , then M is equal to
 (A) 1 (B) 5 (C) 9 (D) 10

[PARAGRAPH TYPE]

Q.11 to Q.15 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Paragraph for question no. 11 & 12

In the given figure, graph of $P(x) = x^4 + ax^3 + bx^2 + cx + d$ is given.



- Q.11 The minimum number of real roots of equation $(P'(x))^2 + P(x)P''(x) = 0$ is equal to
 (A) 3 (B) 4 (C) 5 (D) 6
- Q.12 If the equation $P(x) + k = 0$ has four distinct real roots $\alpha, \beta, \gamma, \delta$ then $[\alpha] + [\beta] + [\gamma] + [\delta]$ is equal to
 [Note : $[t]$ denotes the greatest integer less than or equal to t .]
 (A) -1 (B) -2 (C) 0 (D) 1

Paragraph for question nos. 13 to 15

Let g be a continuous function on $\mathbb{R}$ such that $g(1) = 0$. Also for all $x \in \mathbb{R}$,

$$\int_0^x g(t) dt - \int_x^1 g(t) dt = 2g(x) + 3x + k, \text{ where } k \text{ is some real constant.}$$

- Q.13 The y-intercept of normal drawn to $g(x)$ at $M(1, 0)$ is equal to
 (A) $\frac{2}{3}$ (B) $-\frac{2}{3}$ (C) $\frac{3}{2}$ (D) $-\frac{3}{2}$

- Q.14 Which of the following statement is correct?
 (A) There exist some $c \in \mathbb{R}$ such that $g(x) \geq 2$ for every $x \geq c$.
 (B) $g(x)$ is strictly increasing on $\mathbb{R}$.
 (C) Number of real solutions of the equation $g(x) = 3$ are two.
 (D) There exist some $x_0 \in \mathbb{R}$ such that $g(x_0) = \frac{1}{2}$.

- Q.15 If $\int_0^1 g(x)dx = \lambda(k+3)$ then λ is equal to

- (A) $\frac{1}{2}$ (B) 1 (C) $\frac{1}{3}$ (D) 2

[MULTIPLE CORRECT CHOICE TYPE]

Q.16 to Q.19 has four choices (A), (B), (C), (D) out of which **ONE OR MORE** may be correct.

- Q.16 Let $f(x) = \int_0^x (3t^2 + 2t + 2) dt$, where x satisfies the inequality $\log_2 \left(1 + \sqrt{6x - x^2 - 5} \right) \geq 0$. If minimum and maximum values of $f(x)$ are L and M respectively then
 (A) $L + M = 94$ (B) $L + M = 164$ (C) $M - L = 156$ (D) $M - L = 80$

- Q.17 The curve $y = ax^3 + bx^2 + cx$ is inclined at 45° to x -axis at $(0, 0)$ and it touches x -axis at $(1, 0)$, then
 (A) $f'(1) = 0$ (B) $f''(1) = 6$ (C) $f''(2) = 12$ (D) $f(1) = 0$

- Q.18 Consider the function $f(\theta) = \int_0^1 \frac{\sqrt{1-x^2} - \sin \theta}{\sqrt{1-x^2}} dx$, where $0 \leq \theta \leq \frac{\pi}{2}$, then

- (A) $\max_{0 \leq \theta \leq \frac{\pi}{2}} f(\theta) = 1$ (B) $\min_{0 \leq \theta \leq \frac{\pi}{2}} f(\theta) = \sqrt{2} - 1$
 (C) $\max_{0 \leq \theta \leq \frac{\pi}{2}} f(\theta) = \frac{\pi}{2} - 1$ (D) $\int_0^{\frac{\pi}{2}} f(\theta) d\theta = 2 - \frac{\pi}{2}$

- Q.19 Let $f(x) = \begin{cases} \sin \frac{\pi x}{4}, & 0 \leq x \leq 2 \\ 3 - x, & 2 < x < 4 \\ \{x + 3\}, & 4 \leq x < 5 \end{cases}$

where $\{\cdot\}$ denote fractional part of x . In the interval $(0, 5)$

- (A) global minima does not exist (B) global minima $= -1$
 (C) global maxima does not exist (D) global maxima $= 1$

[INTEGER TYPE]

Q.20 to Q.22 are "Integer Type" questions. (The answer to each of the questions are upto **4 digits**)

Q.20 If $f(x) = \int_x^{x^2} (t-1)dt$, $1 \leq x \leq 2$, then find the global maximum value of $f(x)$.

Q.21 Let f be a continuous function and satisfies $f'(x) > 0$ on $(-\infty, \infty)$ and the value of $f''(x) \forall x \in (0, \infty)$

is equal to minimum value of $\text{Min}_{x \in \mathbb{R}} \{e^{-|x|} + 2, |x| + 2\}$. If $L = \lim_{x \rightarrow \infty} \frac{3x^2 - \frac{3}{x^2 + 1} - 4f'(x)}{f(x)}$ then find the value of $[L^2]$.

[Note : $[k]$ denotes greatest integer less than or equal to k .]

Q.22 Let $M = \left(p, \frac{4}{3-p} - 1\right)$ be a variable point which moves in $x-y$ plane. If $d = \sqrt{a} - \sqrt{b}$, $a, b \in \mathbb{N}$ is the least distance of the point M to the circle $(x-3)^2 + (y+1)^2 = 1$, then find the value of $(a-b)$

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ANSWER KEY

Q.1	B	Q.2	A	Q.3	B	Q.4	B	Q.5	A
Q.6	B	Q.7	A	Q.8	B	Q.9	C	Q.10	D
Q.11	B	Q.12	A	Q.13	B	Q.14	D	Q.15	B
Q.16	BC	Q.17	AD	Q.18	AB	Q.19	AD	Q.20	0004
Q.21	9	Q.22	7						

